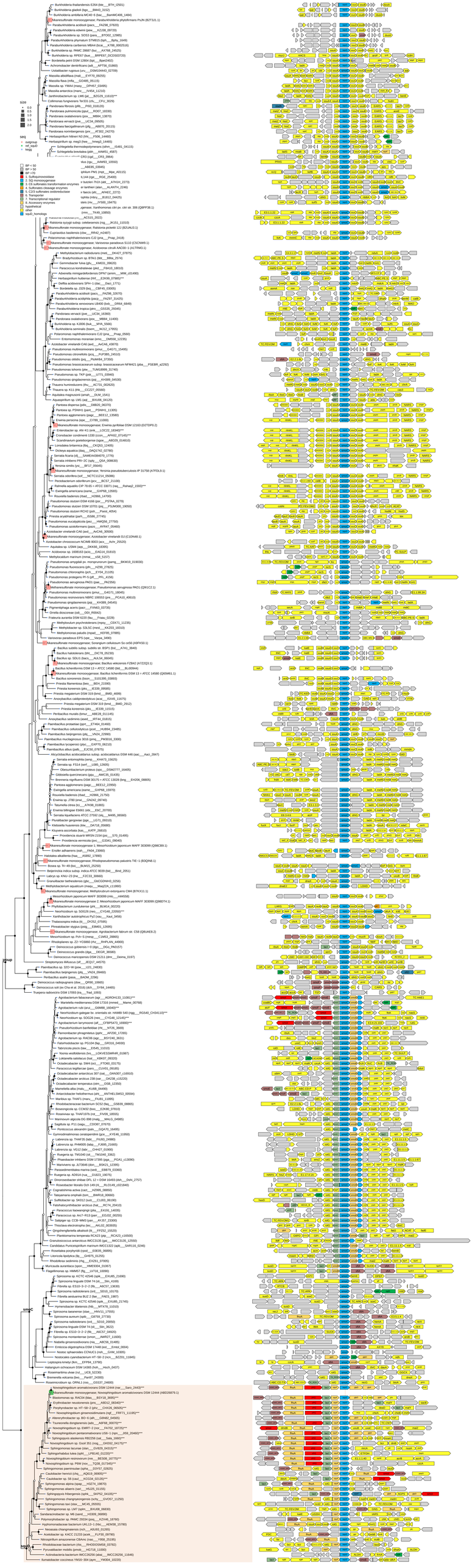

size
0.0
0.5
1.0
1.5
2.0
seq
outgroup
ref_squD
kegg
BP < 50
BP > 50
BP >70
1. Sulfoquinovosidase
2. SQ monooxygenase
3. C6 sulfonates transformation enzymes
4. Sulfonates cleavage enzymes
5. C2/3 sulfonates oxidoreductase
6. Transporter
7. Transcriptional regulator
8. Accessory enzymes
hypothetical
other
squD_homologs
group
smoC
squD
Burkholderia thailandensis E264 (bte___BTH_I2501)
Burkholderia gladioli (bgo___BM43_3152)
Burkholderia ambifaria MC40−6 (bac___BamMC406_1484)
Alkanesulfonate monooxygenase; Paraburkholderia phytofirmans PsJN (B2T3J1.1)
Paraburkholderia acidisoli (pacs___FAZ98_07820)
Paraburkholderia edwinii (pew___KZJ38_09720)
Paraburkholderia sp. SOS3 (para___BTO02_12985)
Paraburkholderia phymatum STM815 (bph___Bphy_1649)
Paraburkholderia caribensis MBA4 (bcai___K788_0002516)
Burkholderia sp. PAMC 28687 (bui___AX768_24525)
Burkholderia sp. RPE67 (bue___BRPE67_DCDS03720)
Bordetella petrii DSM 12804 (bpt___Bpet2402)
Achromobacter denitrificans (adt___APT56_01860)
Usitatibacter rugosus (uru___DSM104443_02709)
Massilia albidiflava (mali___EYF70_09255)
Massilia flava (mfla___GO485_05115)
Massilia sp. YMA4 (masy___DPH57_03495)
Massilia antarctica (manc___IV454_11210)
Janthinobacterium sp. LM6 (jal___BZG29_11610)***
Collimonas fungivorans Ter331 (cfu___CFU_3029)
Pandoraea fibrosis (pfib___PI93_018120)
Pandoraea pulmonicola (ppul___RO07_18330)
Pandoraea oxalativorans (pox___MB84_13870)
Pandoraea vervacti (pve___UC34_09505)
Pandoraea faecigallinarum (pfg___AB870_20115)
Pandoraea norimbergensis (pnr___AT302_24270)
Herbaspirillum hiltneri N3 (hht___F506_14460)
Herbaspirillum sp. meg3 (hee___hmeg3_14465)
Schlegelella thermodepolymerans (sthm___IS481_04115)
Schlegelella brevitalea (pbh___AAW51_4587)
Cupriavidus gilardii CR3 (cgd___CR3_3964)
...ilus (rgu___A4W93_10550)
...___ABE85_03945)
...iphilum PM1 (mpt___Mpe_A0115)
...IL144 (rge___RGE_25400)
...r butzleri 7h1h (abl___A7H1H_2273)
...r lanthieri (alan___ALANTH_2246)
...r faecis (afc___AFAEC_2272)
...yophila (mbry___B1812_04425)
...steis (niv___JY500_19475)
...ygenase; Xanthomonas citri pv. citri str. 306 (Q8PP38.1)
...(rmn___TK49_10850)
Ralstonia insidiosa (rin___ACS15_2022)
Ralstonia syzygii subsp. celebesensis (rsg___JK151_11010)
Alkanesulfonate monooxygenase; Ralstonia pickettii 12J (B2UAU3.1)
Cupriavidus basilensis (cbw___RR42_m2487)
Polaromonas naphthalenivorans CJ2 (pna___Pnap_2418)
Alkanesulfonate monooxygenase; Variovorax paradoxus S110 (C5CNW9.1)
Alkanesulfonate monooxygenase; Acidovorax citrulli AAC00−1 (A1TRM3.1)
Methylobacterium radiodurans (meti___DK427_07875)
Bradyrhizobium sp. BTAi1 (bbt___BBta_2574)
Gemmobacter fulva (gfu___KM031_09620)
Paracoccus kondratievae (pkd___F8A10_18010)
Advenella mimigardefordensis DPN7 (amim___MIM_c01490)
Herbaspirillum huttiense (hhf___E2K99_07985)***
Delftia acidovorans SPH−1 (dac___Daci_1771)
Bordetella sp. J329 (boj___CBF45_03005)
Paraburkholderia acidisoli (pacs___FAZ98_32670)
Paraburkholderia acidiphila (pacp___FAZ97_31425)
Paraburkholderia xenovorans LB400 (bxb___DR64_6849)
Paraburkholderia tropica (ptro___G5S35_29345)
Pandoraea vervacti (pve___UC34_16360)
Pandoraea oxalativorans (pox___MB84_11400)
Burkholderia sp. KJ006 (buk___MYA_5566)
Burkholderia seminalis (bsem___WJ12_17955)
Polaromonas naphthalenivorans CJ2 (pna___Pnap_0560)
Entomomonas moraniae (emo___DM558_12235)
Azotobacter vinelandii CA6 (avd___AvCA6_43870)
Pseudomonas multiresinivorans (pmui___G4G71_15495)
Pseudomonas citronellolis (pcq___PcP3B5_24510)
Pseudomonas otitidis (poj___PtoMrB4_37350)
Pseudomonas brassicacearum subsp. brassicacearum NFM421 (pba___PSEBR_a2292)
Pseudomonas tohonis (ptw___TUM18999_31740)
Pseudomonas sp. TKP (psk___U771_03940)
Pseudomonas qingdaonensis (pqi___KH389_04530)
Thauera humireducens (thu___AC731_002620)
Thauera sp. K11 (thk___CCZ27_05560)
Aquitalea magnusonii (amah___DLM_1541)
Aquaspirillum sp. LM1 (aql___BXU06_04120)
Pantoea dispersa (pdis___D8B20_06370)
Pantoea sp. PSNIH1 (pant___PSNIH1_11305)
Pantoea agglomerans (pagc___BEE12_13580)
Erwinia persicina (epe___CI789_11000)
Alkanesulfonate monooxygenase; Erwinia pyrifoliae DSM 12163 (D2TDP0.2)
Enterobacter sp. AN−K1 (enk___LOC22_18340)***
Cronobacter condimenti 1330 (ccon___AFK62_07145)***
Scandinavium goeteborgense (sgoe___A8O29_014810)
Lonsdalea britannica (lbq___CKQ53_12405)
Dickeya aquatica (daq___DAQ1742_02780)
Serratia ficaria (sfj___SAMEA4384070_1776)
Serratia inhibens PRI−2C (sply___Q5A_008630)
Yersinia similis (ysi___BF17_05645)
Alkanesulfonate monooxygenase; Yersinia pseudotuberculosis IP 31758 (A7FDL9.1)
Serratia odorifera (sof___NCTC11214_05086)
Pectobacterium odoriferum (pcv___BCS7_21100)
Rahnella aquatilis CIP 78.65 = ATCC 33071 (raq___Rahaq2_1593)***
Ewingella americana (eame___GXP68_13565)
Rouxiella badensis (rbad___H2866_14700)
Pseudomonas stutzeri DSM 4166 (psr___PSTAA_0279)
Pseudomonas stutzeri DSM 10701 (psj___PSJM300_19050)
Pseudomonas stutzeri RCH2 (psh___Psest_4054)
Priestia aryabhattai (parh___I5S86_27745)
Pseudomonas eucalypticola (pez___HWQ56_27750)
Pseudomonas azotoformans (pazo___AYR47_05460)
Azotobacter vinelandii CA6 (avd___AvCA6_30500)
Alkanesulfonate monooxygenase; Azotobacter vinelandii DJ (C1DN48.1)
Azotobacter chroococcum NCIMB 8003 (acx___Achr_25520)
Aquitalea sp. USM4 (aqs___DKK66_18395)
Acidovorax sp. 1608163 (acio___EAG14_01810)
Methylocaldum marinum (mmai___sS8_5157)
Pseudomonas amygdali pv. morsprunorum (pamg___BKM19_019030)
Pseudomonas fluorescens (pfn___HZ99_27825)
Pseudomonas chlororaphis (pch___EY04_21105)
Pseudomonas protegens Pf−5 (pfl___PFL_4156)
Pseudomonas aeruginosa PAO1 (pae___PA2356)
Methanesulfonate monooxygenase; Pseudomonas aeruginosa PAO1 (Q9I1C2.1)
Pseudomonas multiresinivorans (pmui___G4G71_18045)
Pseudomonas resinovorans NBRC 106553 (pre___PCA10_40610)
Pseudomonas qingdaonensis (pqi___KH389_04545)
Pigmentiphaga aceris (pacr___FXN63_03735)
Orrella dioscoreae (odi___ODI_R0042)
Frateuria aurantia DSM 6220 (fau___Fraau_0228)
Methylovulum psychrotolerans (mpsy___CEK71_11235)
Methylobacter sp. S3L5C (mesl___KKZ03_18310)
Methylomonas paludis (mpad___KEF85_07885)
Variovorax paradoxus EPS (vpe___Varpa_3490)
Alkanesulfonate monooxygenase; Sorangium cellulosum So ce56 (A9FK50.1)
Bacillus subtilis subsp. subtilis str. BSP1 (bsl___A7A1_3840)
Bacillus halotolerans (bht___DIC78_05230)
Bacillus sp. SDLI1 (bacs___AUL54_06045)
Alkanesulfonate monooxygenase; Bacillus velezensis FZB42 (A7Z2Q3.1)
Bacillus licheniformis DSM 13 = ATCC 14580 (bld___BLi00944)
Alkanesulfonate monooxygenase; Bacillus licheniformis DSM 13 = ATCC 14580 (Q65M61.1)
Bacillus sonorensis (bson___S101395_03893)
Priestia filamentosa (beo___BEH_21590)
Priestia koreensis (pks___IE339_09585)
Priestia megaterium DSM 319 (bmd___BMD_4699)
Anoxybacillus caldiproteolyticus (acai___ISX45_11675)
Priestia megaterium DSM 319 (bmd___BMD_2912)
Priestia koreensis (pks___IE339_11515)
Peribacillus muralis (bmur___ABE28_011145)
Anoxybacillus sediminis (ased___IRT44_01815)
Paenibacillus protaetiae (pprt___ET464_01460)
Paenibacillus cellulosilyticus (pcel___HUB94_23485)
Paenibacillus beijingensis (pbj___VN24_22990)
Paenibacillus mucilaginosus 3016 (pmq___PM3016_3300)
Paenibacillus lycopersici (plyc___GXP70_06210)
Paenibacillus albus (palb___EJC50_07875)
Alicyclobacillus acidocaldarius subsp. acidocaldarius DSM 446 (aac___Aaci_2847)
Serratia entomophila (senp___KHA73_15625)
Serratia sp. FS14 (serf___L085_12605)
Obesumbacterium proteus (opo___DSM2777_16405)
Gibbsiella quercinecans (gqu___AWC35_01435)
Brenneria nigrifluens DSM 30175 = ATCC 13028 (bng___EH206_08805)
Pantoea agglomerans (pagc___BEE12_22950)
Ewingella americana (eame___GXP68_15970)
Rouxiella badensis (rbad___H2866_21750)
Erwinia sp. J780 (erwi___GN242_09740)
Tatumella citrea (tci___A7K98_01865)
Erwinia billingiae Eb661 (ebi___EbC_20700)
Serratia liquefaciens ATCC 27592 (slq___M495_06560)
Pluralibacter gergoviae (pge___LG71_09310)
Klebsiella huaxiensis (klw___DA718_05680)
Kluyvera ascorbata (kas___KATP_26810)
Providencia stuartii MRSN 2154 (psi___S70_01495)
Providencia vermicola (pvc___G3341_09040)
Alkanesulfonate monooxygenase 1; Mesorhizobium japonicum MAFF 303099 (Q98CB9.1)
Ensifer adhaerens (eah___FA04_23060)
Halotalea alkalilenta (haa___A5892_17890)
Alkanesulfonate monooxygenase; Rhodopseudomonas palustris TIE−1 (B3QIN8.1)
Bosea sp. Tri−49 (boi___BLM15_25250)
Beijerinckia indica subsp. indica ATCC 9039 (bid___Bind_2051)
Labrys sp. KNU−23 (lne___FZC33_30660)
Granulibacter bethesdensis (gbc___GbCGDNIH3_0256)
Methylobacterium aquaticum (maqu___Maq22A_c11860)
Alkanesulfonate monooxygenase; Methylorubrum extorquens CM4 (B7KX11.1)
Mesorhizobium japonicum MAFF 303099 (mlo___mll4558)
Alkanesulfonate monooxygenase 2; Mesorhizobium japonicum MAFF 303099 (Q98DT4.1)
Phyllobacterium zundukense (pht___BLM14_00220)
Neorhizobium sp. SOG26 (neo___CYG48_22050)***
Xanthobacter autotrophicus Py2 (xau___Xaut_3456)
Thalassospira indica (tii___DY252_07595)
Phreatobacter stygius (pstg___E8M01_12695)
Alkanesulfonate monooxygenase; Agrobacterium fabrum str. C58 (Q8UAE8.2)
Mesorhizobium sp. Pch−S (mesp___C1M53_28865)
Rhodoplanes sp. Z2−YC6860 (rhz___RHPLAN_44900)
Deinococcus gobiensis I−0 (dgo___DGo_PA0157)
Deinococcus grandis (dga___DEGR_36580)
Deinococcus maricopensis DSM 21211 (dmr___Deima_0197)
Streptomyces liliifuscus (slf___JEQ17_44570)
Paenibacillus sp. 32O−W (pow___IJ21_24830)
Paenibacillus beijingensis (pbj___VN24_09460)
Peribacillus asahii (pasa___BAOM_2296)
Deinococcus radiopugnans (dsw___QR90_10665)
Deinococcus soli (ex Cha et al. 2016) (dch___SY84_14465)
Truepera radiovictrix DSM 17093 (tra___Trad_1093)
Agrobacterium fabacearum (agr___AGROH133_11081)***
Martelella mediterranea DSM 17316 (mmed___Mame_00768)
Agrobacterium rubi (arui___G6M88_16040)***
Neorhizobium galegae bv. orientalis str. HAMBI 540 (ngg___RG540_CH34110)***
Neorhizobium sp. SOG26 (neo___CYG48_12145)***
Agrobacterium larrymoorei (alf___CFBP5473_16900)***
Pseudorhizobium banfieldiae (rht___NT26_3669)
Pannonibacter phragmitetus (pphr___APZ00_17265)
Agrobacterium sp. RAC06 (agc___BSY240_3631)
Falsirhodobacter sp. PG104 (fap___GR316_04930)
Tabrizicola piscis (taw___EI545_11010)
Yoonia vestfoldensis (lvs___LOKVESSMR4R_01987)
Loktanella salsilacus (lsal___KBK07_09320)
Octadecabacter sp. SW4 (oct___FTO60_03175)
Paracoccus tegillarcae (paro___CUV01_09185)
Octadecabacter antarcticus 307 (oat___OAN307_c16910)
Octadecabacter arcticus 238 (oar___OA238_c15220)
Octadecabacter temperatus (otm___OSB_12350)
Mameliella alba (malu___KU6B_04490)
Antarctobacter heliothermus (aht___ANTHELSMS3_00594)
Maribius sp. THAF1 (maru___FIU81_11895)
Rhodobacteraceae bacterium SC52 (faq___G5B39_08865)
Boseongicola sp. CCM32 (boo___E2K80_07810)
Roseivivax sp. THAF197b (rot___FIV09_18555)
Marinovum algicola DG 898 (malg___MALG_04985)
Sagittula sp. P11 (sagu___CDO87_07670)
Ponticoccus alexandrii (palx___GQA70_16495)
Gymnodinialimonas ceratoperidinii (gce___KYE46_11950)
Labrenzia sp. THAF35 (labt___FIU93_24980)
Labrenzia sp. PHM005 (labp___FJ695_21665)
Labrenzia sp. VG12 (labr___CHH27_01900)
Ruegeria sp. TM1040 (sit___TM1040_3362)
Phaeobacter inhibens DSM 17395 (pga___PGA1_c13090)
Marivivens sp. JLT3646 (don___BSK21_12395)
Parasedimentitalea marina (sedi___EBB79_03360)
Ruegeria sp. AD91A (rua___D1823_19075)
Dinoroseobacter shibae DFL 12 = DSM 16493 (dsh___Dshi_2757)
Roseobacter litoralis Och 149 (rli___RLO149_c021840)
Cognatishimia activa (cact___HZ995_08850)
Tateyamaria omphalii (tom___BWR18_00660)
Sulfitobacter sp. SK012 (sulz___C1J03_06190)
Falsihalocynthiibacter arcticus (hat___RC74_20410)
Paracoccus liaowanqingii (plia___E4191_14035)
Paracoccus sp. Arc7−R13 (parr___EOJ32_00255)
Salipiger sp. CCB−MM3 (yan___AYJ57_23300)
Thioclava electrotropha (tec___AKL02_003935)
Qingshengfaniella alkalisoli (lit___FPZ52_15520)
Planktomarina temperata RCA23 (ptp___RCA23_c16500)
Granulosicoccus antarcticus IMCC3135 (gai___IMCC3135_12550)
Candidatus Puniceispirillum marinum IMCC1322 (apb___SAR116_0246)
Roseitalea porphyridii (rpod___E0E05_06895)
Litoricola lipolytica (llp___GH975_01255)
Rhodoferax sediminis (rhg___EXZ61_07005)
Muricauda aurantiaca (spon___HME9304_01067)
Flagellimonas sp. HMM57 (flg___LV716_10090)
Spirosoma sp. KCTC 42546 (spik___EXU85_21690)
Spirosoma linguale DSM 74 (sli___Slin_4169)
Fibrella sp. ES10−3−2−2 (fib___A6C57_13630)
Spirosoma radiotolerans (srd___SD10_10170)
Fibrella aestuarina BUZ 2 (fae___FAES_1987)
Spirosoma sp. KCTC 42546 (spik___EXU85_21745)
Hymenobacter tibetensis (htb___MTX78_11010)
Spirosoma taeanense (stae___HNV11_17555)
Spirosoma aureum (spib___G8759_27730)
Spirosoma radiotolerans (srd___SD10_20050)
Spirosoma linguale DSM 74 (sli___Slin_3622)
Fibrella sp. ES10−3−2−2 (fib___A6C57_04920)
Spirosoma montaniterrae (smon___AWR27_11600)
Niabella ginsenosidivorans (nia___A8C56_01485)
Emticicia oligotrophica DSM 17448 (eol___Emtol_0004)
Nostoc sphaeroides CCNUC1 (nsh___GXM_10305)
Nostocales cyanobacterium HT−58−2 (ncn___BZZ01_11845)
Leptospira kmetyi (lkm___EFP84_13790)
Haliangium ochraceum DSM 14365 (hoh___Hoch_0437)
Roseimaritima ulvae (rul___UC8_52230)
Bremerella volcania (bvo___Pan97_24300)
Roseimicrobium sp. ORNL1 (roo___G5S37_24600)
Novosphingobium aromaticivorans DSM 12444 (nar___Saro_2443)***
Alkanesulfonate monooxygenase; Novosphingobium aromaticivorans DSM 12444 (ABD26879.1)
Blastomonas sp. RAC04 (blas___BSY18_3695)***
Erythrobacter neustonensis (pns___A9D12_08340)***
Porphyrobacter sp. HT−58−2 (phz___CHX26_06505)***
Novosphingobium ginsenosidimutans (ngf___FRF71_11195)***
Altererythrobacter sp. BO−6 (alh___G6N82_04565)
Tsuneonella dongtanensis (ado___A6F68_00070)***
Novosphingobium sp. EMRT−2 (nor___FA702_18725)***
Novosphingobium pentaromativorans US6−1 (npn___JI59_20460)***
Sphingopyxis alaskensis RB2256 (sal___Sala_1660)***
Novosphingobium sp. Gsoil 351 (nog___GKE62_04175)***
Sphingomonas lacunae (slan___GV829_04315)***
Sphingorhabdus lutea (sphl___LPB140_01220)***
Novosphingobium resinovorum (nre___BES08_16775)***
Novosphingobium sp. P6W (nov___TQ38_017345)***
Sphingomonas paeninsulae (spha___D3Y57_02825)
Caulobacter henricii (chq___AQ619_06905)***
Caulobacter sp. S6 (caul___KCG34_02135)***
Sphingomonas alpina (spap___H3Z74_19870)
Sphingomonas aliaeris (sari___H5J25_01155)
Sphingopyxis fribergensis (sphk___SKP52_04105)***
Sphingomonas changnyeongensis (schy___GVO57_11250)
Sphingomonas taxi (stax___MC45_05555)
Sphingomonas sp. LM7 (splm___BXU08_06830)
Sandaracinobacter sp. M6 (sand___H3309_06890)
Polymorphobacter sp. PAMC 29334 (poq___KZX46_18760)
Hyphomonadaceae bacterium UKL13−1 (hbc___AEM38_15780)
Neoasaia chiangmaiensis (nch___A0U93_01265)
Acetobacter sp. KACC 21233 (acek___FLP30_09790)
Nitrospirillum amazonense CBAmc (nao___Y958_25195)
Rhodobiaceae bacterium (rbs___RHODOSMS8_03782)
Pyruvatibacter mobilis (pmob___HG718_11000)
Actinobacteria bacterium IMCC26256 (abai___IMCC26256_11646)
Ilumatobacter coccineus YM16−304 (aym___YM304_10220)